



Simulated Display Outputs

“Escape from Behind those Bars”

Retro hardware fans like us are typically faced with a dichotomy. We LOVE the look and feel of our old displays, because when we have them tuned *just right*, our games and demos become more like a time-machine, teleporting us back to the most nostalgic places in our imaginations.

However, we also know that old hardware can be a royal pain in the “*you know what*” to manage and maintain, as it degrades over time. One of those challenges comes in the form of video distortion that can proliferate from dirty power sources, leaking signals in the PCB, external EMF, etc., etc.

Over the years, various techniques have been developed that have helped to reduce some of these effects, such as eliminating the old RF Bypass circuit, installing a lumafix, implementing video amplifiers, etc.

One of the most notable solutions came from Jay Desmarais (aka [Jaystonian](#)) with his ClearVideo circuit. He took the idea of using a high quality video amp (the THS7316), with an enhanced Lumafix, then went much further, by providing the ability to selectively inject inverted versions of the most common noise sources back into the video signal (e.g., the RAS, CAS, PHI, etc.).

EVO's ClearVideo64 Circuit and VICII Controls

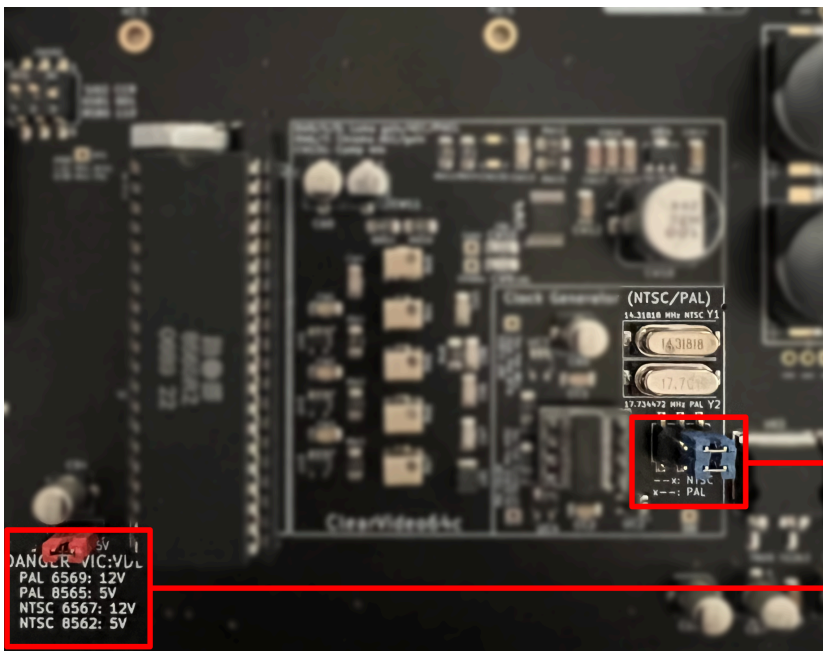
EVO incorporates a *modified* version of the ClearVideo circuit, which includes some extra features and niceties, such as additional tuning adjustments, which help to reduce noise even further.

EVO provides a series of jumper headers that allow you to select the required voltage levels for the VIC-II chip(s) that you intend to use*:

Voltage	PAL VIC-II Chips
12v	MOS 6569 – (PAL-B, used in most PAL countries)
12v	MOS 6572 – (PAL-N, used in southern South America only)
12v	MOS 6573 – (PAL-M, used in Brazil only)
5v	MOS 8565 – HMOS-II version for "C64E" motherboards
Voltage	NTSC VIC-II Chips
12v	MOS 6567 – Original NMOS version
5v	MOS 8562 – HMOS-II version

EVO also includes an onboard clock generator circuit, which provides a rock solid and jitter free voltage controlled oscillator (VCO) signal. This design was inspired by the original work done by [Jeff Birt \(aka Hey Birt\)](#), although EVO takes this things a bit further, by enabling you to selectively activate either a PAL or NTSC timing crystal, so you can enjoy a wider selection of software titles.

The table below highlights the areas on the EVO board where VIC-II voltages and PAL/NTSC timings can be setup*



A PAL / NTSC Switching*

Separate jumpers are here for the Dot clock & the PHI color clock.
**Be sure to move both jumpers*

MODE	1	2	3
NTSC	ON	ON	off
PAL	off	ON	ON

B VIC-II Voltage Config*

MODE	12V	5V
NTSC	6567	8562
PAL	6569	8565

***CAUTION* Supplying 12V to a 5V VIC-II is likely to fry it**

***IMPORTANT NOTES:**

- Always reference the table above when changing your VIC-II chip, to ensure that you're providing the **correct** voltage.
- ***Passing too much voltage can completely destroy your VICII chip***
- Passing too low of a voltage won't allow it to work
- Always ensure that your system is **powered off** before attempting to change any voltages

NOTE for Kawari VIC-II Replacement Users:

The [Kawari](#) by Randy Rossi is a wonderful replacement option for the aging VIC-II chips, which are becoming more difficult to find and more expensive to source, with the likelihood that they will eventually fail increasing over time.

You'll be happy to know that we have successfully tested both the Kawari-Large and the Kawari-Mini with EVO, in both PAL and NTSC modes, while using either 40 and 80 column output. In our testing experience, the video output looks absolutely terrific, with accurate and stable screen raster timings.

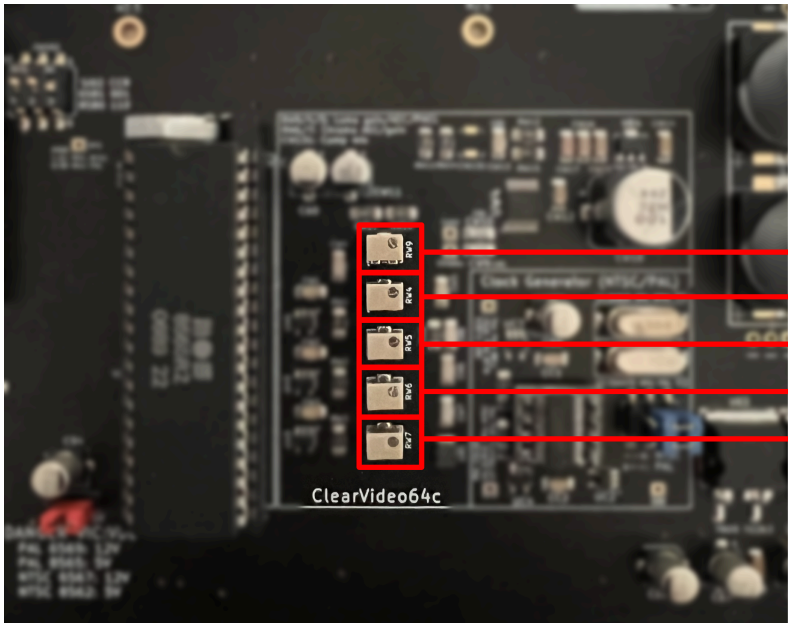
1. Be sure to let Randy know that you intend to use your Kawari with EVO, since he will want to provide you with an SRAM compatible "core" (firmware code) that will ensure the best results (as of this writing, the v1.14 core is tested to work well with EVO in both PAL & NTSC video modes)
2. Referencing the **VIC-II Voltage Config** settings in the above diagram, you should configure your EVO board to provide 5V to your Kawari

Tuning your Video Output for Perfection

The type and level of video distortion that your machine and monitor are producing will depend on a number of key factors, including input power sources, internal voltage regulator, the age and types of VIC-II chip(s) you are using, the stability, speed and jitter of your color clock and VCO, leakage from signals that are located near each other on the circuit board, as well as the types of displays you're using (e.g., a CRT versus an LCD through a video upscaler, etc.).

Getting your video image “dialed in just right” is an **iterative process**. It will take a bit of time and patience to get it fully worked out, for the best possible image quality, but you'll thank yourself when it's done.

The table below highlights the areas on the EVO board where ClearVideo64c is located, highlighting the adjustable potentiometers that allow for gain level attenuation and noise reduction.

	A RW9 - Luma Gain Luminosity signal level adjustments
	B RW4 - Luma AEC Reduce noise from Address Enable Control
	C RW5 - Luma PHCL Reduce noise from Color Clock
	D RW6 - Choma AEC Reduce noise from Address Enable Control
	E RW7- Chroma Gain Chroma signal level adjustment

Eliminate those “Jail Bars” and “Tune Away” Video Noise

- Use a small **1.4mm flathead screwdriver** to adjust all of the potentiometers clockwise, until they emit a faint clicking sound. You will need to rotate in a full 360 degrees, *multiple times*. They include a built-in protection that prevents over-rotation or damage
- Start by adjusting the middle potentiometer (**C RW5 - Luma PHCL**), turning it counter-clockwise, until you can see a ~50% overlap in video
- Now move up to the next pot (**B RW4 - Luma AEC**) and adjust it slowly, until the picture starts to improve a bit more. Jail bars should be starting to fade a bit now
- Move up to **A RW9 - Luma Gain** and adjust to taste in the same way, until the color level looks right
- Move down to **D RW6 - Choma AEC**
- Then finally to **E RW7- Chroma Gain**
- Now, go back and repeat the process, with smaller increments, beginning with RW5, then RW4, then RW9, etc., adjusting slowly, in +/- 2 turns, then move down to RW6 & RW7

Continue this iterative process until you've achieved *your* desired video output quality. There's no right or wrong settings here, *it's completely up to you...*

Examples of common C64 display issues, due to various types of signal noise:



With EVO's ClearVideo circuit, a couple of potentiometer twists can tune away any remnant noise or distortion, while boosting your color saturation and enhancing your text clarity. Best of all, you'll finally be able to *break free* from those ugly jail bars!

Here's some of the feedback we've been hearing:

*"The best way I can describe this is that it looks **fake**. It's like somebody connected an emulator up to my monitor instead of a real C64."* - **David Murraray (The 8 Bit guy)**, Feb 1st 2023

*"After my very brief setup process, we got rid of most of the jail bars... there's barely **any**... and usually they are pretty visible especially if you use modern screens..."* - **Jan Beta**, Feb 4th 2023

*"The audio and video is just engineered to an exacting standard here, it's **amazing**"*
- **Matt Demicco (Retrobits)**, Dec 29th 2022

Audio / Video Cables (AV)

Most of the standard AV cables made for the C64 are not going to provide support for a second SID's audio output. This leaves you with a few choices to consider.

OPTION 1: You can buy a more modern AV cable from a supplier like [8 Bit Classics](#), which supports second-SID audio through Pin-7 of the AV jack (note: we are not affiliated with or compensated by them)



Standard A/V Cable

S-Video

Composite Video

Stereo Audio Out

OPTION 2: You can pick up a nice breakout module from the [Retrospective shop](#), which can plug directly into your system's AV Port and provide separate output signals for the left and right audio channels, as well as separate signals for composite and S-Video. This approach enables you to use standard, off the shelf RCA and S-Video cables, which are widely available.

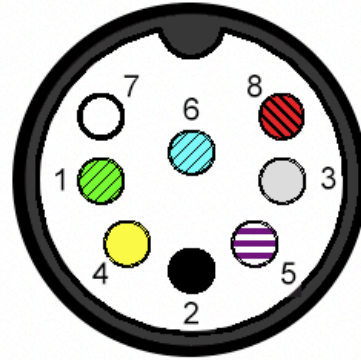


OPTION 3: You can have your own AV cable(s) made, by following the specification below, but you'll need to be mindful of the details. The C64's 8-Pin DIN uses a "horseshoe shaped" configuration at a 262 degree angle. There are other 8-Pin connector standards with pins on different angles, *which won't fit your machine properly!*

DIN-8 262



Female (Computer)



Male (Cable)

- 1. S-Video Luma
- 2. Ground
- 3. Audio Right Output (Red RCA jack center)
- 4. Composite (Yellow RCA jack center)
- 5. Audio Input to SID #1
- 6. S-Video Chroma
- 7. Audio Left Output (White RCA jack center)
- 8. +5V